

From Direct Algebraic Generating Functions to Elliptic Sigma and Theta Representations of Somos–4 Sequences

A Constructive Workflow

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Abstract

A direct five-parameter algebraic generating function can be constructed whose Hankel transform follows a prescribed nondegenerate Somos–4 orbit. Its discriminant contains the genus-one curve

$$E_J: \quad y^2 = (X^2 - JX + \tau)^2 - 4X.$$

The elliptic-function theory behind Somos–4 sequences is classical; the practical difficulty lies in moving from concrete recurrence data to a numerical sigma- or theta-function representation without losing track of normalizations. The relevant ingredients are scattered across several sources and conventions. This paper brings them together into one constructive and reproducible workflow:

$$(\lambda, m, r, \eta, \tau) \mapsto J \mapsto E_J \mapsto (g_2, g_3) \mapsto (\omega_1, \omega_2) \mapsto (z_0, \delta) \mapsto \sigma \mapsto \theta_1.$$

The emphasis is on normalization choices, computational checks, and an executable PARI/GP implementation. No new proof of the classical sigma-function representation is intended; the contribution is a coherent end-to-end procedure that starts from the algebraic generating function and ends with a verified sigma or theta description.

1 Introduction

The relation between Somos–4 recurrences, elliptic curves, and elliptic functions is classical; see, in particular, [5, 2, 4, 3]. Generic Somos–4 orbits admit representations in terms of the Weierstrass sigma function [2]. Nevertheless, a reader beginning with a concrete sequence still faces a substantial practical problem: the relevant constructions occur in different normalizations, coordinate systems, and parametrizations, and the intermediate steps needed for a numerical implementation are rarely presented together.

The purpose of this paper is not to prove a new elliptic-function theorem. Rather, it is to make the established theory directly usable. Our starting point is the direct algebraic generating function constructed in [1]. Its discriminant naturally produces a genus-one curve associated with the Somos–4 data. From there we describe how to obtain a Weierstrass model, compute the period lattice, identify the translation point, determine the Abel–Jacobi parameters, and finally write explicit sigma- and theta-function representations. The continued-fraction and Hankel viewpoint is closely related to the genus-one construction in [6].

The paper is organized as a mathematical workflow rather than a theorem–proof sequence. Classical facts are stated with references and are not reproved. The main concern is to keep the conventions aligned and to expose the checks that catch a wrong branch, period, sign, or scaling. PARI/GP fragments accompany the construction, and the appendix contains a complete verification script.

1.1 Scope and contribution

This paper provides:

- one route from the five input parameters to an elliptic curve;
- an explicit record of the normalization choices needed to pass from a quartic model to sigma and theta functions;
- computational checkpoints at each stage;
- worked examples comparing recurrence and sigma-function evaluations, together with a fully explicit theta conversion; and
- an end-to-end PARI/GP workflow suitable for adaptation.

It does not claim a new proof of the classical Somos–4 sigma formula, a new integrability result, or a new theorem on elliptic curves.

2 Input data and the direct algebraic generating function

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be the algebraic generating function constructed in [1]. It satisfies

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0,$$

where the polynomials A, B, C depend explicitly on $(\lambda, m, r, \eta, \tau)$. Its associated Hankel transform is initialized by

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta,$$

and satisfies

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2. \quad (1)$$

Throughout the main workflow we assume that all displayed denominators and sigma factors are nonzero and that the elliptic discriminant is nonzero. Exceptional cases are interpreted on the projective curve or by continuation in the parameters.

The generating-function discriminant factors as

$$B(z)^2 - 4A(z)C(z) = (\lambda^2 \eta m r^2 z)^2 \left((1 - Jz + \tau z^2)^2 - 4z^3 \right),$$

where

$$J = \frac{\lambda \eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda \eta}. \quad (2)$$

Removing the square prefactor and applying the reciprocal change of variable suggested by the residual polynomial gives the quartic model

$$E_J : \quad y^2 = (X^2 - JX + \tau)^2 - 4X. \quad (3)$$

The subsequent workflow begins with equations (2) and (3).

3 What information is already contained in the discriminant?

The factorization above provides the conceptual bridge between the algebraic and transcendental descriptions. Once the irrelevant square factor is removed, the generating-function discriminant supplies a quartic genus-one model depending only on (J, τ) . Thus the five input parameters determine the initial placement and normalization of the orbit, while the curve itself is encoded by the reduced pair (J, τ) .

This observation should be interpreted carefully. The discriminant already contains the elliptic curve, but it does not by itself fix every choice needed for a numerical theta representation. One must still:

1. choose and verify a birational map to a Weierstrass model;
2. adopt an ordering and orientation for the periods;
3. identify the point implementing one Somos step;
4. choose Abel–Jacobi lifts of the initial point and translation;
5. determine the elementary exponential factors; and
6. check the result against the original recurrence.

In this precise sense, the theta function is hidden in the discriminant, but extracting it requires a controlled sequence of normalizations.

4 The constructive workflow

4.1 Workflow Step 1: Compute the invariant and construct the quartic

Given $(\lambda, m, r, \eta, \tau)$, compute J using equation (2) and form the quartic equation (3). Before proceeding, test that the quartic is nonsingular. Equivalently, its polynomial discriminant must be nonzero. Degenerate inputs require a separate rational or singular analysis and are outside the principal workflow.

Checkpoint. Expand the two sides of the generating-function discriminant identity and verify exact equality in the coefficient field.

4.2 Workflow Step 2: Convert the quartic to Weierstrass form

Choose as origin the point at infinity characterized by $y/X^2 \rightarrow 1$. Put

$$U = \frac{y + X^2 - JX + \tau}{2}, \quad V = U(2X - J) - 1. \quad (4)$$

Since the quartic equation implies

$$U(U - (X^2 - JX + \tau)) = -X,$$

a direct elimination of X and y yields

$$V^2 = 4U^3 + (J^2 - 4\tau)U^2 + 2JU + 1. \quad (5)$$

The shift

$$u = U + \frac{J^2 - 4\tau}{12}, \quad v = V \quad (6)$$

therefore gives the Weierstrass model

$$v^2 = 4u^3 - g_2u - g_3, \quad (7)$$

where, with $a = J^2 - 4\tau$,

$$g_2 = \frac{a^2}{12} - 2J, \quad g_3 = -\frac{a^3}{216} + \frac{aJ}{6} - 1. \quad (8)$$

The inverse affine formulas are equally simple. First set

$$U = u - \frac{J^2 - 4\tau}{12}, \quad V = v.$$

For $U \neq 0$,

$$X = \frac{V + JU + 1}{2U}, \quad y = 2U - X^2 + JX - \tau. \quad (9)$$

Substitution verifies both compositions identically on their affine charts. The denominator U vanishes at the two cubic points $(U, V) = (0, 1)$ and $(0, -1)$. In the projective correspondence these are two points omitted by the affine inverse chart. The chosen origin $y/X^2 \rightarrow 1$ maps to the cubic point at infinity. The other quartic point at infinity maps to $(U, V) = (0, 1)$, while $(U, V) = (0, -1)$ maps to the finite quartic point $(X, y) = (0, -\tau)$. Thus the sole affine exceptional set of the inverse formula is $U = 0$; it is resolved by using the projective models.

The polynomial discriminant of the quartic in X is

$$\text{disc}_X((X^2 - JX + \tau)^2 - 4X) = 256\Delta, \quad (10)$$

where

$$\Delta = g_2^3 - 27g_3^2 = J^4\tau + J^3 - 8J^2\tau^2 - 36J\tau + 16\tau^3 - 27. \quad (11)$$

Consequently, over a field of characteristic zero, the birational elliptic model is nonsingular precisely when $\Delta \neq 0$; no additional parameter values are excluded.

Checkpoint. Reduce

$$V^2 - (4U^3 + (J^2 - 4\tau)U^2 + 2JU + 1)$$

modulo $y^2 - (X^2 - JX + \tau)^2 + 4X$ and verify that the remainder is zero. Verify the two inverse compositions, equation (10), and

$$j = 1728 \frac{g_2^3}{\Delta}.$$

4.3 Workflow Step 3: Compute and normalize the period lattice

Let e_1, e_2, e_3 be the distinct roots of

$$4u^3 - g_2u - g_3 = 0.$$

They are the branch values of the double covering

$$v^2 = 4(u - e_1)(u - e_2)(u - e_3).$$

A period lattice is obtained by integrating the holomorphic differential du/v over an oriented homology basis:

$$2\omega_1 = \oint_{\gamma_1} \frac{du}{v}, \quad 2\omega_2 = \oint_{\gamma_2} \frac{du}{v}.$$

The cycles are oriented and, if necessary, replaced by an equivalent basis so that

$$\operatorname{Im} \left(\frac{\omega_2}{\omega_1} \right) > 0.$$

We then set $\tau_{\text{mod}} = \omega_2/\omega_1$. Only the lattice is intrinsic; the ordered pair $(2\omega_1, 2\omega_2)$ depends on the chosen homology basis. This choice must therefore be retained unchanged in the later definitions of elliptic logarithms, sigma functions, and theta functions.

For real input data with $\Delta > 0$, the three roots are real and may be ordered as $e_1 > e_2 > e_3$. Define

$$k^2 = \frac{e_2 - e_3}{e_1 - e_3}, \quad k'^2 = 1 - k^2 = \frac{e_1 - e_2}{e_1 - e_3}.$$

With the positive real square root of $e_1 - e_3$, a convenient rectangular half-period basis is

$$\omega_1 = \frac{K(k)}{\sqrt{e_1 - e_3}}, \quad \omega_2 = \frac{iK(k')}{\sqrt{e_1 - e_3}}, \quad (12)$$

where

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Consequently, $\tau_{\text{mod}} = iK(k')/K(k)$ lies in the upper half-plane.

For general complex parameters, or for real parameters with only one real root, it is preferable to compute the periods from integrals over a chosen pair of branch cuts rather than to impose the ordering used in equation (12). The resulting basis is then normalized by an element of $\operatorname{SL}_2(\mathbb{Z})$ so that $\operatorname{Im}(\tau_{\text{mod}}) > 0$. A direct elliptic-curve period routine provides an independent numerical check [9, 8].

Checkpoint. Verify

$$4(u - e_1)(u - e_2)(u - e_3) = 4u^3 - g_2u - g_3, \quad \operatorname{Im}(\tau_{\text{mod}}) > 0.$$

Then reconstruct the invariants from the normalized lattice and check that they agree with the original g_2 and g_3 to the working precision. In the three-real-root case, also compare equation (12) with direct numerical integration of du/v .

4.4 Workflow Step 4: Determine the elliptic translation point

Introduce the gauge-invariant recurrence variables

$$x_n = \frac{H_{n-1}H_{n+1}}{H_n^2}. \quad (13)$$

The Somos–4 recurrence equation (1) implies the second-order QRT map

$$x_{n+2}x_nx_{n+1}^2 = x_{n+1} + \tau, \quad x_{n+2} = \frac{x_{n+1} + \tau}{x_nx_{n+1}^2}, \quad (14)$$

and the quantity

$$x_nx_{n+1} + \frac{1}{x_n} + \frac{1}{x_{n+1}} + \frac{\tau}{x_nx_{n+1}} \quad (15)$$

is independent of n . Its value is precisely J from equation (2). Indeed,

$$x_2 = \frac{\lambda r}{m^2}, \quad x_3 = \frac{m\eta}{r^2},$$

and substituting these values into equation (15) gives equation (2).

The invariant biquadratic becomes the quartic equation (3) under

$$X_n = x_n x_{n+1}, \quad y_n = x_n - x_{n+1}. \quad (16)$$

The sign convention in equation (16) is essential: replacing y_n by $-y_n$ reverses the direction of the translation point below. Applying equations (4) and (6) gives

$$\begin{aligned} U_n &= \frac{y_n + X_n^2 - JX_n + \tau}{2}, & V_n &= U_n(2X_n - J) - 1, \\ u_n &= U_n + \frac{J^2 - 4\tau}{12}, & Q_n &= (u_n, V_n) \in E_J. \end{aligned} \quad (17)$$

With the group law on equation (7) and the point at infinity as identity, one QRT step is

$$Q_{n+1} = Q_n + P, \quad P = \left(\frac{J^2 - 4\tau}{12}, 1 \right). \quad (18)$$

Thus the translation point is determined directly by (J, τ) and need not be recovered numerically from several iterates. A direct check constructs Q_{n+1} from equations (14) and (16) and ?? and verifies $Q_{n+1} - Q_n = P$ by the Weierstrass group law. The formula is valid on the open chart where the displayed expressions are defined; exceptional or vanishing recurrence values are handled projectively or by continuation.

There is also a continued-fraction interpretation. On the quartic model, the two points at infinity differ by a fixed divisor class. Advancing one line of the genus-one continued fraction translates the Jacobian by that class [6]. With the origin chosen in Workflow Step 2, the second point at infinity maps to $(U, V) = (0, 1)$, hence to P in equation (18). The QRT and continued-fraction routes therefore give the same increment.

For the classical data $H_1 = H_2 = H_3 = H_4 = 1$ and $\tau = 1$, one has $J = 4$ and

$$E_J: \quad v^2 = 4u^3 - 4u + 1, \quad P = (1, 1).$$

Furthermore, $x_2 = x_3 = 1$ and equation (14) gives $x_4 = 2$. The corresponding points are

$$Q_2 = (0, 1), \quad Q_3 = (-1, -1),$$

and the group law gives $Q_3 - Q_2 = (1, 1) = P$.

Checkpoint. For several consecutive pairs (x_n, x_{n+1}) , verify equation (15), map the states to Q_n , and check $Q_{n+1} - Q_n = P$. As a quick structural test, confirm directly that P lies on equation (7); in the intermediate coordinates it is $(U, V) = (0, 1)$.

4.5 Workflow Step 5: Compute Abel–Jacobi coordinates

Choose lifts z_2 and δ in $\mathbb{C}$ satisfying

$$(\wp(z_2), \wp'(z_2)) = Q_2, \quad (\wp(\delta), \wp'(\delta)) = P,$$

or, equivalently,

$$z_2 = \int_{\infty}^{Q_2} \frac{du}{v}, \quad \delta = \int_{\infty}^P \frac{du}{v} \pmod{\Lambda}.$$

Then set

$$z_0 = z_2 - 3\delta \pmod{\Lambda}. \quad (19)$$

The lifts are not unique. Their representatives should be reduced to a stated fundamental parallelogram, with signs fixed by the v -coordinates. With the normalization used in the companion script, the geometric point indexed by Q_n is represented by

$$z_n = z_0 + (n+1)\delta,$$

so that $Q_{n+1} - Q_n = P$ is encoded by the increment δ in the Abel–Jacobi parameter.

Checkpoint. For several integers n , verify that

$$(\wp(z_0 + (n+1)\delta), \wp'(z_0 + (n+1)\delta))$$

agrees with the Weierstrass image of Q_n .

4.6 Workflow Step 6: Fix the sigma-function normalization

In the convention used here, the classical sigma-function solution of a nondegenerate Somos–4 recurrence is

$$H_n = AB^n \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}. \quad (20)$$

This is the normalization of the general sigma representation proved in [2]. The sign of δ is fixed in Workflow Steps 4 and 5 by requiring $Q_{n+1} = Q_n + P$ and $(\wp(\delta), \wp'(\delta)) = P$. With the convention used here, the Abel–Jacobi coordinate of Q_n is

$$z_n = z_0 + (n+1)\delta, \quad (\wp(z_n), \wp'(z_n)) = Q_n. \quad (21)$$

In particular, the initial recurrence data determine Q_2 from x_2 and x_3 , so z_0 is obtained from equation (19). This shift is essential: using z_2 itself as z_0 would displace the sigma formula by three recurrence indices.

For brevity, put

$$S_n = \frac{\sigma(z_0 + n\delta)}{\sigma(\delta)^{n^2}}. \quad (22)$$

Assuming the displayed sigma factors are nonzero, $H_1 = \lambda$ and $H_2 = m$ determine the gauge constants uniquely:

$$B = \frac{mS_1}{\lambda S_2}, \quad A = \frac{\lambda^2 S_2}{mS_1^2}. \quad (23)$$

Equivalently,

$$B = \frac{m \sigma(z_0 + \delta) \sigma(\delta)^3}{\lambda \sigma(z_0 + 2\delta)}, \quad (24)$$

$$A = \frac{\lambda^2 \sigma(z_0 + 2\delta)}{m \sigma(\delta)^2 \sigma(z_0 + \delta)^2}. \quad (25)$$

The remaining initial values are consistency conditions:

$$r = AB^3 S_3, \quad \eta = AB^4 S_4. \quad (26)$$

Thus all four initial values are used: the first two fix A and B , while the third and fourth test the geometric data and branch choices.

The exponent n^2 in equation (20) is fixed by the sigma addition formula

$$\frac{\sigma(z + \delta) \sigma(z - \delta)}{\sigma(z)^2 \sigma(\delta)^2} = \wp(\delta) - \wp(z). \quad (27)$$

Substitution of equation (20) into equation (13) gives

$$x_n = \frac{\sigma(z_0 + (n-1)\delta)\sigma(z_0 + (n+1)\delta)}{\sigma(z_0 + n\delta)^2\sigma(\delta)^2} = \wp(\delta) - \wp(z_0 + n\delta). \quad (28)$$

This identity links the sigma normalization directly to the QRT variables.

Checkpoint. First verify equation (26). Then compare equation (20) with exact recurrence values for a range of positive and, where defined, negative indices. Independently check equation (28); this detects an incorrect sign of δ , an index shift in z_0 , or an inconsistent period representative.

4.7 Workflow Step 7: Convert the sigma formula to theta form

For the chosen ordered half-periods, define

$$\tau_{\text{mod}} = \frac{\omega_2}{\omega_1}, \quad \eta_1 = \zeta(\omega_1), \quad \vartheta'_1 = \theta'_1(0 \mid \tau_{\text{mod}}), \quad c = \frac{2\omega_1}{\pi\vartheta'_1}.$$

The classical sigma–theta identity is

$$\sigma(z) = \exp\left(\frac{\eta_1 z^2}{2\omega_1}\right) c \theta_1\left(\frac{\pi z}{2\omega_1} \mid \tau_{\text{mod}}\right), \quad (29)$$

with the standard analytic normalization of θ_1 ; see [7, 8]. Put

$$A_0 = \frac{\pi\delta}{2\omega_1}, \quad B_0 = \frac{\pi z_0}{2\omega_1}. \quad (30)$$

Substituting equation (29) into equation (20) and collecting the constant, linear, and quadratic powers of n gives the explicit theta form

$$H_n = K_0 q_0^{n^2} r_0^n \theta_1(A_0 n + B_0 \mid \tau_{\text{mod}}), \quad (31)$$

where

$$K_0 = A c \exp\left(\frac{\eta_1 z_0^2}{2\omega_1}\right), \quad (32)$$

$$q_0 = \exp\left(\frac{\eta_1 \delta^2}{2\omega_1}\right) \sigma(\delta)^{-1} = (c \theta_1(A_0 \mid \tau_{\text{mod}}))^{-1}, \quad (33)$$

$$r_0 = B \exp\left(\frac{\eta_1 z_0 \delta}{\omega_1}\right). \quad (34)$$

The exponential in equation (33) is essential. Equivalently, applying equation (29) at $z = \delta$ gives the second expression for q_0 .

Checkpoint. Evaluate equation (20) and equation (31) independently and compare them before comparing either expression with the recurrence. Software libraries often use different argument scalings or nomes for Jacobi theta functions; therefore equations (29) and (30) should be treated as the normalization contract for any implementation.

5 Algorithm summary

- Step 1.** Read $(\lambda, m, r, \eta, \tau)$ and test the required nonvanishing assumptions.
- Step 2.** Compute J from equation (2).
- Step 3.** Construct E_J and test nonsingularity.
- Step 4.** Compute the birational Weierstrass model and retain both maps.
- Step 5.** Compute (g_2, g_3) and check curve invariants.
- Step 6.** Compute (ω_1, ω_2) and normalize the period basis.
- Step 7.** Construct consecutive QRT states and verify the fixed point P .
- Step 8.** Compute and normalize z_0 and δ .
- Step 9.** Determine A and B from the initial data and check H_3, H_4 .
- Step 10.** Convert the result to theta form using equations (29), (30) and (32) to (34).
- Step 11.** Validate recurrence, generating-function discriminant, sigma form, and theta form.

6 Numerical examples

6.1 The classical Somos–4 sequence

For

$$(\lambda, m, r, \eta, \tau) = (1, 1, 1, 1, 1),$$

the companion script computes

$$J = 4, \quad (g_2, g_3, \Delta) = (4, -1, 37).$$

The corresponding quartic is

$$y^2 = (X^2 - 4X + 1)^2 - 4X.$$

One compatible pair of half-periods, obtained at 80 decimal digits of working precision and oriented so that $\text{Im}(\omega_2/\omega_1) > 0$, is

$$\omega_1 = 1.4967293231159798149, \quad \omega_2 = 1.2256946909933950304i.$$

The corresponding Abel–Jacobi parameters are

$$z_0 = -2.7887781458561870232 + 1.2256946909933950304i, \quad \delta = 1.8591854305707913488.$$

The sigma-normalization constants are

$$A = -0.23674493915977148 + 0.06797467161582982i,$$

$$B = 1.24599348339577469 + 3.55401276446268772i.$$

Although these constants and lifts are complex, their combination in equation (20) gives real integral values to numerical precision. The result is

$$H_1, H_2, \dots = 1, 1, 1, 1, 2, 3, 7, 23, 59, \dots,$$

the classical Somos–4 sequence, OEIS A006720 [10].

6.2 A nontrivial member of the five-parameter family

For $(\lambda, m, r, \eta, \tau) = (2, 1, 1, 3, 2)$, exact arithmetic gives

$$J = \frac{43}{6}, \quad (g_2, g_3, \Delta) = \left(\frac{2213809}{15552}, -\frac{3291850729}{10077696}, \frac{1161689}{324} \right).$$

The first terms of the orbit are

$$2, 1, 1, 3, \frac{5}{2}, \frac{41}{2}, 74, \frac{2051}{6}, \frac{43103}{6}, \frac{335987}{9}, \dots$$

This nonunit rational seed shows that the exact algebraic part of the workflow remains consistent away from the symmetric seed $(1, 1, 1, 1, 1)$ before the numerical period and sigma evaluation is performed.

7 Computational reliability and reproducibility

The workflow mixes exact algebra and high-precision complex analysis. The following practices are recommended:

- perform discriminant factorization and birational-map checks exactly whenever the input data are rational;
- postpone floating-point conversion until period computation;
- state whether reported periods are full periods or half-periods;
- state the orientation and period-basis convention;
- reduce Abel–Jacobi arguments consistently modulo the same lattice;
- test signs with the second Weierstrass coordinate, not only the u -coordinate;
- compare the sigma and theta representations independently; and
- validate several consecutive and nonconsecutive indices.

These checks are not cosmetic. A wrong sign of δ , a shifted lift of z_0 , or a modularly equivalent but inconsistently transformed period basis can produce formulas that appear plausible for a few terms while failing globally.

8 Discussion and future directions

The discriminant of the direct algebraic generating function provides a particularly transparent entry point into the elliptic geometry of a Somos–4 orbit. The present workflow is intended to make that geometry computationally accessible rather than to re-establish the classical elliptic-function theory. It also clarifies which parts of the transcendental representation are intrinsic to the curve and which record the initial point, indexing, and normalization.

Natural extensions include:

- direct extraction of the translation point from paired-Hankel or continued-fraction data;
- systematic formulas for modular invariants in terms of (J, τ) ;
- arithmetic analysis of special parameter choices;
- analogous workflows for Somos–5 and Somos–6; and
- higher-genus generalizations involving Kleinian sigma functions.

9 Conclusion

The elliptic curve needed for a sigma- or theta-function description is already present in the discriminant of the direct algebraic generating function. What remains is a sequence of explicit, verifiable conversions. By collecting those conversions, fixing their conventions, and implementing them in one script, the scattered classical theory becomes a usable tool for concrete Somos-4 sequences.

A PARI/GP companion implementation

The code below carries the normalization used in the text through the exact algebraic checks, the period calculation, the QRT translation check, and the elliptic sigma evaluation. PARI/GP represents

$$v^2 = 4u^3 - g_2u - g_3$$

by

$$Y^2 = u^3 - \frac{g_2}{4}u - \frac{g_3}{4}, \quad Y = \frac{v}{2}.$$

Thus a text point (u, v) is supplied to PARI/GP as $[u, v/2]$. The functions used below are documented in [9].

```
\\ PARI/GP companion for the algebraic-to-sigma workflow.

somos_J(lambda, m, r, eta, tau) =
{
  lambda*eta/(m*r)
  + m^2/(lambda*r)
  + r^2/(m*eta)
  + tau*m*r/(lambda*eta);
};

quartic_poly(J, tau, X) = (X^2 - J*X + tau)^2 - 4*X;

qrt_invariant(x, y, tau) = x*y + 1/x + 1/y + tau/(x*y);
qrt_step(x, y, tau) = [y, (y + tau)/(x*y^2)];
qrt_step_pair(x, y, tau) = qrt_step(x, y, tau);

somos_recurrence_step(v, tau) =
{
  my(n = #v);
  if (n < 4, error("Need at least four values to advance the Somos-4 orbit"));
  if (v[n-3] == 0, error("Zero denominator in Somos-4 recurrence"));
  (v[n]*v[n-2] + tau*v[n-1]^2) / v[n-3];
};

somos_orbit(lambda, m, r, eta, tau, N) =
{
  my(v = List([lambda, m, r, eta]));
  if (N < 1, return([]));
  for (n = 5, N,
    listput(v, somos_recurrence_step(v, tau));
  );
  Vec(v)[1..min(N, #v)];
};

quartic_to_weierstrass(X, Y, J, tau) =
```

```

{
  my(U = (Y + X^2 - J*X + tau)/2);
  my(V = U*(2*X - J) - 1);
  [U + (J^2 - 4*tau)/12, V];
};

weierstrass_to_quartic(u, v, J, tau) =
{
  my(U = u - (J^2 - 4*tau)/12);
  if (U == 0, error("Affine inverse map hits the excluded U = 0 chart"));
  my(X = (v + J*U + 1)/(2*U));
  [X, 2*U - X^2 + J*X - tau];
};

weierstrass_invariants(J, tau) =
{
  my(a = J^2 - 4*tau);
  my(g2 = a^2/12 - 2*J);
  my(g3 = -a^3/216 + a*J/6 - 1);
  [g2, g3, g2^3 - 27*g3^2];
};

ell_curve_from_J(J, tau) =
{
  my(w = weierstrass_invariants(J, tau));
  ellinit([0, 0, 0, -w[1]/4, -w[2]/4]);
};

fixed_translation_point(J, tau) =
{
  my(a = J^2 - 4*tau);
  [a/12, 1/2];      \\ PARI coordinates [u,v/2]
};

translation_point_on_curve(J, tau) = fixed_translation_point(J, tau);

curve_point_from_qrt(x, y, J, tau) =
{
  my(P = quartic_to_weierstrass(x*y, x-y, J, tau));
  [P[1], P[2]/2];    \\ PARI coordinates [u,v/2]
};

check_discriminant(J, tau) =
{
  my(w = weierstrass_invariants(J, tau));
  my(F = quartic_poly(J, tau, 'X));
  [poldisc(F, 'X), 256*w[3], poldisc(F, 'X) - 256*w[3]];
};

check_quartic_weierstrass(J, tau) =
{
  my(w = weierstrass_invariants(J, tau));
  my(E = ell_curve_from_J(J, tau));
  my(W = ellperiods(E));
  my(omega1 = W[1]/2, omega2 = -W[2]/2);
  my(tau_mod = omega2/omega1);
};

```

```

print("J = ", J, "; g2 = ", w[1], "; g3 = ", w[2], "; Delta = ", w[3]);
print("PARI full periods W = ", W);
print("Reference half-periods: omega1 = ", omega1, ", omega2 = ", omega2);
print("tau_mod = ", tau_mod);
print("discriminant check = ", check_discriminant(J, tau));

if (imag(tau_mod) <= 0,
    print("Warning: tau_mod is not in the upper half-plane.");
);

abs(check_discriminant(J, tau)[3]) < 1e-30;
};

verify_qrt_translation(lambda, m, r, eta, tau, nsteps = 6) =
{
    my(J = somos_J(lambda, m, r, eta, tau));
    my(E = ell_curve_from_J(J, tau));
    my(x = lambda*r/m^2, y = m*eta/r^2);
    my(P = fixed_translation_point(J, tau));
    for (k = 1, nsteps,
        my(Qn = curve_point_from_qrt(x, y, J, tau));
        my(next = qrt_step(x, y, tau));
        my(Qnp1 = curve_point_from_qrt(next[1], next[2], J, tau));
        print("step ", k,
            ": invariant error = ", qrt_invariant(x, y, tau) - J,
            ", translation exact = ", ellsub(E, Qnp1, Qn) == P);
        x = next[1];
        y = next[2];
    );
};

sigma_lattice_from_J(J, tau) =
{
    my(E = ell_curve_from_J(J, tau));
    ellperiods(E);
};

sigma_representation(lambda, m, r, eta, tau, z0, delta) =
{
    my(L = sigma_lattice_from_J(somos_J(lambda, m, r, eta, tau), tau));
    my(sd = ellsigma(L, delta));
    my(S1 = ellsigma(L, z0 + delta));
    my(S2 = ellsigma(L, z0 + 2*delta));
    my(B = m*S1*sd^3 / (lambda*S2));
    my(A = lambda^2*S2 / (m*sd^2*S1^2));
    [A, B, sd, S1, S2];
};

sigma_parameters(lambda, m, r, eta, tau) =
{
    my(J = somos_J(lambda, m, r, eta, tau));
    my(E = ell_curve_from_J(J, tau));
    my(x2 = lambda*r/m^2, x3 = m*eta/r^2);
    my(Q2 = curve_point_from_qrt(x2, x3, J, tau));
    my(P = fixed_translation_point(J, tau));
    my(L = ellperiods(E));
    my(delta = ellpointtoz(E, P));
};

```

```

my(z2 = ellpointtoz(E, Q2));
\\ In the H_1-started normalization, Q_n corresponds to z0 + (n+1)*delta.
my(z0 = z2 - 3*delta);
my(sd = ellsigma(L, delta));
my(S1 = ellsigma(L, z0 + delta)/sd);
my(S2 = ellsigma(L, z0 + 2*delta)/sd^4);
my(B = m*S1/(lambda*S2));
my(A = lambda^2*S2/(m*S1^2));
[E, L, z0, delta, A, B];
};

sigma_value(sigpar, n) =
{
  my(L = sigpar[2], z0 = sigpar[3], delta = sigpar[4]);
  my(A = sigpar[5], B = sigpar[6]);
  A*B^n*ellsigma(L, z0 + n*delta)/ellsigma(L, delta)^(n^2);
};

somos_sigma_value(lambda, m, r, eta, tau, z0, delta, n) =
{
  my(L = sigma_lattice_from_J(somos_J(lambda, m, r, eta, tau), tau));
  my(A, B, sd, S1, S2);
  [A, B, sd, S1, S2] = sigma_representation(lambda, m, r, eta, tau, z0, delta);
  A*B^n*ellsigma(L, z0 + n*delta)/sd^(n^2);
};

workflow_demo(lambda, m, r, eta, tau, N = 10, prec = 80) =
{
  default(realprecision, prec);
  my(J = somos_J(lambda, m, r, eta, tau));
  my(w = weierstrass_invariants(J, tau));
  if (w[3] == 0, error("Singular quartic / elliptic curve"));
  my(sigpar = sigma_parameters(lambda, m, r, eta, tau));
  my(exact = somos_orbit(lambda, m, r, eta, tau, N));
  my(approx = vector(N, n, sigma_value(sigpar, n)));
  my(err = vector(N, n, abs(approx[n] - exact[n])));
  print("J = ", J);
  print("[g2,g3,Delta] = ", w);
  print("ellperiods(E) = ", ellperiods(sigpar[1]));
  print("z0 = ", sigpar[3], "; delta = ", sigpar[4]);
  print("A = ", sigpar[5], "; B = ", sigpar[6]);
  print("exact orbit = ", exact);
  print("sigma orbit = ", approx);
  print("absolute errors = ", err);
  [J, w, ellperiods(sigpar[1]), sigpar, exact, approx, err];
};

demo_sigma_workflow() =
{
  workflow_demo(1, 1, 1, 1, 1, 8, 80);
};

check_sigma_workflow_classical() =
{
  workflow_demo(1, 1, 1, 1, 1, 10, 80);
};

```

```

classical_example() =
{
  workflow_demo(1, 1, 1, 1, 1, 10, 80);
};

nonunit_example() =
{
  workflow_demo(2, 1, 1, 3, 2, 10, 80);
};

```

The script deliberately performs the recurrence and sigma evaluations through separate code paths. The displayed error vector is therefore a direct check of the indexing, elliptic logarithms, period convention, and gauge factors. The theta representation can be checked independently in software that implements the standard $\theta_1(z \mid \tau)$ convention by using equations (30) and (32) to (34).

B Verification checklist

1. The generating-function discriminant factorization is exact.
2. The quartic is nonsingular.
3. Forward and inverse birational maps are verified symbolically.
4. Curve discriminants and j -invariants agree under normalization.
5. The period basis has $\text{Im}(\tau_{\text{mod}}) > 0$.
6. The computed point difference $Q_{n+1} - Q_n$ is independent of n .
7. The signs of elliptic logarithms agree with the v -coordinates.
8. Sigma values satisfy all four initial conditions and the recurrence.
9. Theta values agree independently with sigma values.
10. Reported numerical errors are compatible with the working precision.

C Statement on AI assistance and computational tools

This manuscript was prepared with AI-assisted language editing, L^AT_EX formatting, and development of verification code. The author reviewed the mathematical statements, normalization choices, references, and numerical outputs. PARI/GP was used for exact algebra and high-precision elliptic computations; its relevant conventions are documented in [9].

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